

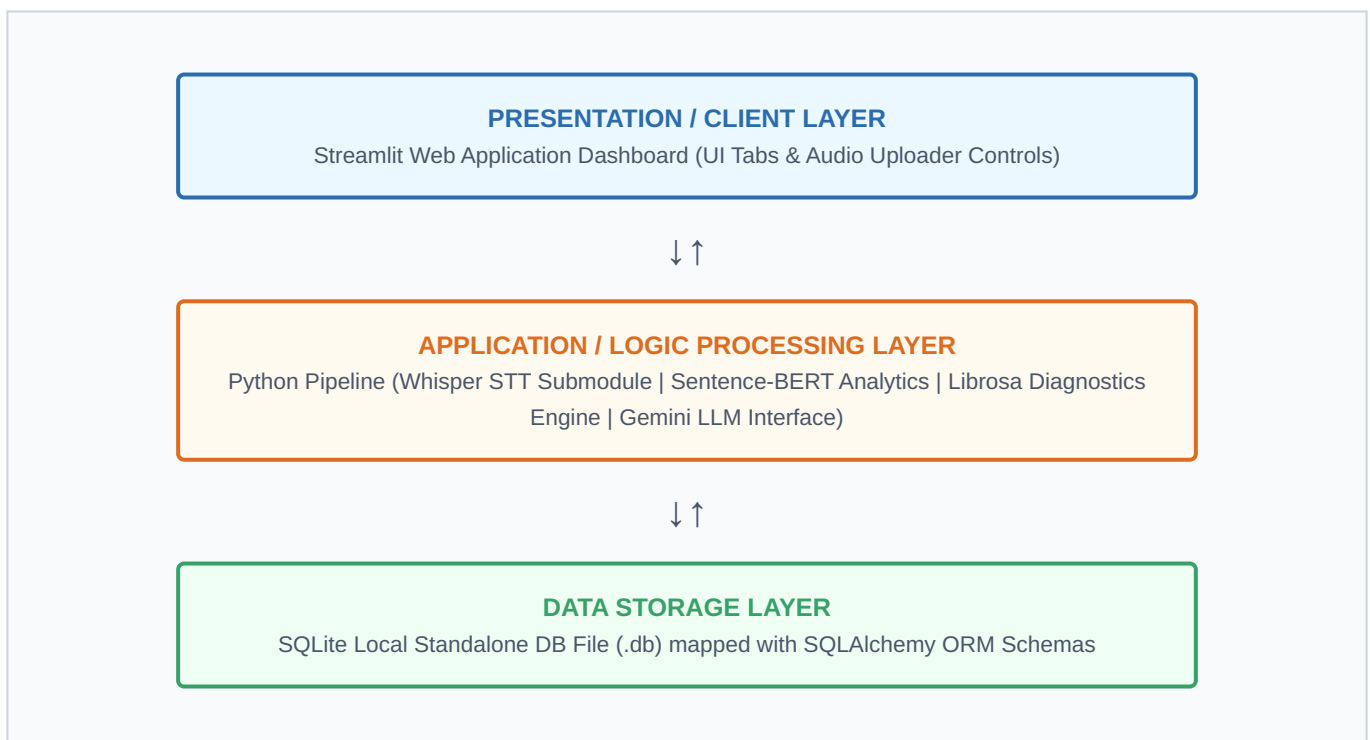
Date	15 March 2026
Team ID	XXXXXX
Project Name	VBCUA — Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

SOLUTION ARCHITECTURE

SOLUTION ARCHITECTURE DIAGRAM:

Provide a high-level visual representation of your solution's architecture mapped into Presentation Layer, Application Layer, and Data Layer with your specific components.

HIGH-LEVEL ARCHITECTURE BLUEPRINT



COMPONENT DESCRIPTION TABLE

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the multi-tab browser dashboard for the evaluation interface, handles voice explanation file upload actions, and dynamically displays the analysis metrics alongside generated graphs and report triggers.	Streamlit Framework, Pandas, Matplotlib Waveform Rendering
Speech Transcription	Handles backend file stream translation inputs by loading localized speech weights to extract accurate verbal string sequences from uploaded speech media files.	OpenAI Whisper (base model size)
Semantic Scoring	Computes vector embeddings for both the raw transcribed verbal output and the verified conceptual topic benchmark text to extract an objective structural understanding metric.	Sentence-Transformers (all-MiniLM-L6-v2)
Fluency Analytics	Tracks raw signal characteristics to analyze pause lengths, identify silence frequencies, and cross-reference localized phrase tokens to enumerate speaking rate parameters (WPM tracking).	Librosa acoustic processor, NumPy, Soundfile tracking subroutines
Critique Generator	Orchestrates composite scoring matrix values to synthesize structured natural language commentary pointing out user concept gaps and delivery guidance templates.	Google Gemini API (gemini-1.5-flash with localized backup engine)
Data Infrastructure	Manages transaction workflows, persists historical records, maps object relationships safely, and lets users look up past performance evaluations on the dashboard.	SQLite local file database storage engine backend, SQLAlchemy ORM mappings